

Lab: Derivation Exercise — Precision Optimization and Attention

Objective

Derive the mathematical equations governing precision estimation and show how precision optimization implements attention in hierarchical models.

Part 1: Univariate Precision Estimation

Goal: Derive the optimal precision for the simplest Gaussian model.

Model: $p(o|\mu) = N(o; \mu, \pi^{-1})$, with known mean μ and unknown precision π . Prior: $p(\pi) = \text{Gamma}(\pi; \alpha_0, \beta_0)$.

Task:

1. Write the log-likelihood $\ln p(o_{1:N} | \mu, \pi)$ for N observations
2. Combine with the log-prior $\ln p(\pi) = (\alpha_0 - 1)\ln \pi - \beta_0\pi + \text{const}$
3. Find the posterior $p(\pi | o_{1:N})$ by recognizing it as a Gamma distribution with updated parameters
4. Show that the posterior mean $E[\pi | o_{1:N}]$ is related to the inverse sample variance

Write your response here

Part 2: Precision in Hierarchical Predictive Coding

Goal: Derive how precision modulates message passing at each level.

Model: Two-level hierarchy with precision parameters π_1 (observation level) and π_2 (state level):

- $p(o|s_1) = N(o; s_1, \pi_1^{-1})$
- $p(s_1|s_2) = N(s_1; s_2, \pi_2^{-1})$

Task:

1. Write the free energy F as a function of sufficient statistics μ_1, μ_2 and precision parameters π_1, π_2
2. Compute $\partial F / \partial \mu_1$ and show it depends on the ratio π_1 / π_2 — the relative precision of sensory and prior prediction errors
3. Show that when $\pi_1 \gg \pi_2$ (high sensory precision), updates are driven primarily by sensory data; when $\pi_2 \gg \pi_1$ (high prior precision), updates are driven primarily by priors

Write your response here

Part 3: Log-Precision Dynamics

Goal: Derive the gradient flow for log-precision $\gamma = \ln \pi$.

Task:

1. Express free energy as a function of γ (not π directly)
2. Compute $\partial F / \partial \gamma = \partial F / \partial \pi \cdot \partial \pi / \partial \gamma = \partial F / \partial \pi \cdot \pi$
3. Show that the dynamics $d\gamma/dt = -\partial F / \partial \gamma$ stably estimate precision from prediction error statistics
4. Explain why using log-precision (which ranges over all reals) is numerically more stable than raw precision (which must be positive)

Write your response here

Part 4: Fisher Information and Confidence

Goal: Connect precision to the information content of observations.

Task:

1. Compute the Fisher information $I(\theta) = -E[\partial^2 \ln p(o|\theta)/\partial \theta^2]$ for $p(o|\theta) = N(\theta, \pi^{-1})$
2. Show that $I(\theta) = \pi$ — higher precision observations carry more information about hidden states
3. State the Cramér-Rao bound: $\text{Var}(\theta) \geq 1/I(\theta)$. Explain what this means for inference under different precision regimes
4. Discuss the implication: attention (increasing π) $\rightarrow$ more Fisher information $\rightarrow$ tighter Cramér-Rao bound $\rightarrow$ more confident inference

Write your response here

Part 5: Synthesis

In 200 words, explain how precision serves as the mathematical bridge between free energy theory (abstract variational inference) and attention (concrete cognitive process). Connect the derivations from Parts 1-4 into a unified account.

Write your response here

Lab Summary

Part	Skill Developed	Mathematical Result
1	Bayesian estimation	Conjugate precision estimation (Gamma-Gaussian)
2	Hierarchical analysis	Precision ratio modulates sensory vs. prior influence
3	Dynamical systems	Log-precision gradient flow
4	Information geometry	Fisher information = precision; Cramér-Rao bound
5	Conceptual integration	Precision as the math of attention